

# **SUMMER PRACTICE REPORT**

**Course Code:** [Course Code]

**Student Name:** [Your Name Surname]

**Student ID:** [Your Student ID]

**Company Name:** TÜBİTAK BİLGEM

**Division/Branch:** System Integration and Test Department

**Internship Dates:** [Start Date] - [End Date]

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# 1 Description of the Company

TÜBİTAK BİLGEM (Informatics and Information Security Research Center) is a national research and development center in Turkey. It operates under the Scientific and Technological Research Council of Turkey (TÜBİTAK). The company focuses on information security, telecommunications, software development, and testing of various defense and civilian systems.

The center employs over 2000 personnel, the majority of whom are engineers and researchers holding advanced degrees. The administrative structure consists of several institutes and departments, one of which is the System Integration and Test Department where this summer practice was conducted. This department is specifically responsible for the integration of complex subsystems and conducting rigorous testing, especially on Radio Frequency (RF) systems, to ensure reliability and performance. The end products are generally strategic defense systems, communication devices, and cryptographic equipment used by governmental and military organizations.

## 2 A Brief History of the Company

TÜBİTAK BİLGEM was established to merge several existing research institutes to create a unified body capable of tackling large-scale, multidisciplinary projects. Over the years, BİLGEM has played a pivotal role in national technological independence by developing indigenous cryptographic algorithms, secure communication systems, and advanced radar and electronic warfare systems. Its successful delivery of critical projects has solidified its position as the leading R&D center in Turkey's defense and information technology sectors.

## 3 Introduction

The main purpose of this summer practice was to gain practical experience in the testing and integration of Radio Frequency (RF) systems. TÜBİTAK BİLGEM was chosen due to its advanced laboratory facilities and its central role in developing cutting-edge RF technologies. During the practice, theoretical research and practical test script development were carried out. The primary focus was on understanding RF components, specifically Low Noise Amplifiers (LNAs), and developing automated test routines using Vector Network Analyzers (VNAs). The following chapters detail the theoretical knowledge acquired and the automated testing software developed to interact with the test equipment.

## 4 Work Conducted at the SP Location

During the first days of the internship, the primary focus was on familiarizing myself with the System Integration and Test department. The work mainly revolves around the testing stage of various RF systems. Several research topics were assigned, including RF components (Local Oscillators, Low Noise Amplifiers, Mixers, Filters, Splitters, and Power Dividers), LVDS message packaging, and test equipment (Signal Generators, Spectrum Analyzers, and Vector Network Analyzers).

## 4.1 Theoretical Background on RF Systems

A fundamental concept in RF design is that the length of an antenna is proportional to the wavelength of the signal. This necessitates high-frequency operation for practical antenna sizes, typically around a quarter of the wavelength. Other advantages of high-frequency operation include efficiency in propagation and immunity to certain forms of noise.

Noise is a critical factor in RF systems. Various types of noise exist, such as white noise which affects all frequencies equally, and pink noise which has a power density that falls with increasing frequency. In electronic circuits, thermal noise, shot noise, and flicker noise are common, while phase noise is particularly significant in RF applications [1].

## 4.2 RF Modules and Transceivers

An RF module is a small electronic circuit used to receive, transmit, or transceive radio waves. A transmitter produces radio waves by applying a radio frequency current to an antenna. Conversely, a receiver captures radio waves via an antenna and converts the information into a usable form. A transceiver combines both transmitter and receiver functionalities.

The RF front end is a crucial part of a receiver, encompassing all circuitry between the antenna and the mixer stage. It processes the signal at the original incoming radio frequency before down-conversion to an intermediate frequency. A simple block diagram of a transceiver system can be visualized in Figure 1.

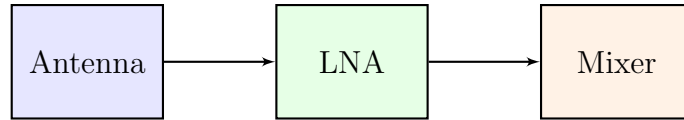


Figure 1: Simple block diagram of an RF transceiver system front-end.

## 4.3 Low Noise Amplifier (LNA) Theory

The LNA is the first active block in a receiver chain, located immediately after the antenna. According to the Friis formula, the noise figure of this first stage dominates the overall noise performance of the receiver [2]. Therefore, the primary function of the LNA is to amplify weak signals while introducing the minimum possible electronic noise.

Designing an LNA requires balancing several conflicting parameters:

- **Noise Figure vs. Impedance Matching:** The optimum source impedance for minimal noise ( $Z_{opt}$ ) rarely aligns with the complex conjugate impedance needed for maximum power transfer ( $Z_{in}^*$ ). Designers use Smith Charts and noise circles to find an acceptable compromise.
- **Stability:** An amplifier can potentially oscillate. Unconditional stability is achieved when input and output reflection coefficients are less than 1, and the Linvill stability factor ( $K$ ) is greater than 1.
- **Linearity:** The amplifier must handle strong interfering signals without distortion. Key metrics include the 1-dB Compression Point ( $P_{1dB}$ ) and the Third-Order Intercept Point ( $IP_3$ ). High linearity often requires higher power consumption.

- **Forward Gain ( $S_{21}$ ):** Sufficient gain is needed to suppress the noise of subsequent stages, but excessive gain can saturate the mixer.

Table 1 summarizes the main requirements for an ideal LNA.

Table 1: Primary requirements for a Low Noise Amplifier.

| Requirement      | Description                                 |
|------------------|---------------------------------------------|
| Low Noise Figure | Minimal added noise from the device.        |
| High Linearity   | Better $IP_3$ to handle strong signals.     |
| Low Power        | Efficient power consumption.                |
| Adequate Gain    | Sufficient forward gain ( $S_{21}$ ).       |
| Matching         | Impedance matched to the antenna or filter. |

## 4.4 Automated VNA Testing for LNA

To practically apply these concepts, an automated test script was developed in Python to interface with a Vector Network Analyzer (VNA) using the VISA protocol. The objective was to measure the forward gain ( $S_{21}$ ) of an LNA across a specified frequency range (2 GHz to 3 GHz). The source code for this automated test script is provided in Appendix A.

### 4.4.1 Test Equipment Connections

Testing an LNA requires careful setup using specific RF instruments. The two primary test setups utilize a Vector Network Analyzer (VNA), a Signal Generator (SG), and a Spectrum Analyzer (SA).

**1. VNA Test Setup (S-Parameters & Gain):** The VNA is used to measure the scattering parameters, such as Gain ( $S_{21}$ ) and Return Loss ( $S_{11}$ ,  $S_{22}$ ). The VNA Port 1 is connected to the LNA Input, and Port 2 is connected to the LNA Output.

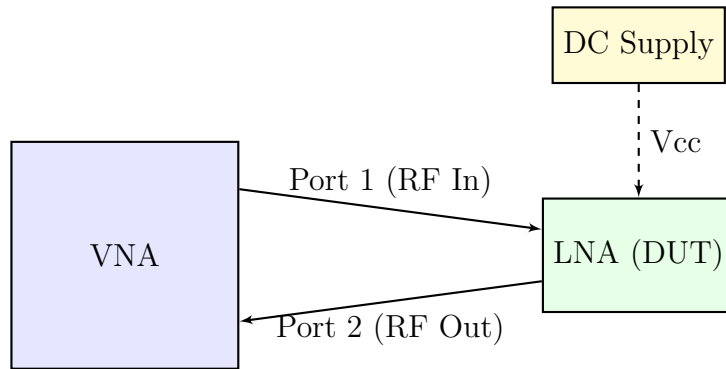


Figure 2: Schematic of LNA Test Setup using a Vector Network Analyzer (VNA).

**2. Signal Generator & Spectrum Analyzer Setup (Linearity & Harmonics):** To measure the 1-dB compression point ( $P_{1dB}$ ) or harmonic distortion, a Signal Generator provides a single-tone or two-tone input, and the Spectrum Analyzer measures the output power.

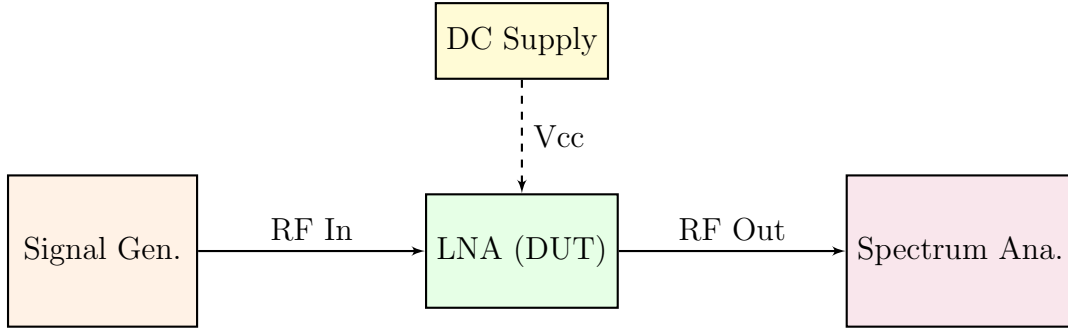


Figure 3: Schematic of LNA Test Setup using a Signal Generator and Spectrum Analyzer.

The script initializes the VISA resource manager, connects to the VNA, and configures the instrument for an  $S_{21}$  measurement. It triggers a sweep, retrieves the frequency and magnitude data, and then plots the results using Matplotlib. Figure 4 represents an example output plot that highlights the maximum gain achieved by the LNA.



Figure 4: Sample plot of LNA Gain ( $S_{21}$ ) vs Frequency obtained from the automated VNA test.

## 5 Conclusions

During this summer practice at TÜBİTAK BİLGEM, significant knowledge was gained regarding the testing and theoretical foundations of RF systems. Understanding the trade-offs in LNA design, such as balancing noise figure, gain, linearity, and stability, provided a deep insight into practical RF engineering. Furthermore, developing automated test scripts using Python to control complex test instruments like the VNA proved to be an invaluable skill.

The rigorous engineering culture at TÜBİTAK BİLGEM demonstrated the importance of precision and reliability in critical systems. This experience successfully bridged the gap between theoretical academic knowledge and practical industry applications. I highly recommend this department to future students interested in RF electronics and automated test systems.

## References

- [1] "LNA (Low Noise Amplifier) basics," YouTube video. Accessed: June 24, 2026. [Online]. Available: <https://www.youtube.com/watch?v=azgW5qCxwcQ>
- [2] H. T. Friis, "Noise Figures of Radio Receivers," *Proceedings of the IRE*, vol. 32, no. 7, pp. 419-422, July 1944.

## A Automated VNA Test Script

The following Python script was developed to automate the VNA for measuring the  $S_{21}$  parameter of an LNA.

```
1 import pyvisa
2 import numpy as np
3 import matplotlib.pyplot as plt
4 import time
5
6 # =====
7 # TUBITAK LNA Automated Test Script (SG + SA)
8 # Measures Scalar Gain ( $|S_{21}|$ ) over frequency.
9 # Note:  $S_{11}$ ,  $S_{22}$ , and K-factor require a VNA (phase measurement).
10 # This script synchronizes a Signal Generator and Spectrum Analyzer.
11 # =====
12
13 def run_sg_sa_test():
14     rm = pyvisa.ResourceManager()
15
16     print("Scanning for connected instruments...")
17     resources = rm.list_resources()
18     print(f"Found: {resources}")
19
20     if len(resources) < 2:
21         print("Need at least 2 instruments (SG and SA) connected. Exiting...")
22         return
23
24     sg_address = resources[0]
25     sa_address = resources[1]
26
27     try:
28         # Open connections
29         sg = rm.open_resource(sg_address)
30         sa = rm.open_resource(sa_address)
31
32         sg.timeout = 5000
33         sa.timeout = 5000
34
35         print(f"Connected to SG: {sg.query('*IDN?').strip()}")
36         print(f"Connected to SA: {sa.query('*IDN?').strip()}")
37
38         # Test Parameters
39         start_freq = 2.0e9 # 2 GHz
40         stop_freq = 3.0e9 # 3 GHz
41         num_points = 51
42         input_power_dbm = -30.0 # Keep input power low to avoid LNA saturation
43
44         frequencies = np.linspace(start_freq, stop_freq, num_points)
45         gains = []
46
47         # Configure Spectrum Analyzer
48         sa.write('*RST')
49         sa.write(':SENS:FREQ:SPAN 5e6') # 5 MHz span around center frequency
```



```

50     sa.write(':DISP:WIND:TRAC:Y:RLEV 0 DBM') # Reference level 0
dBm
51     sa.write(':INIT:CONT OFF') # Single sweep mode
52
53     # Configure Signal Generator
54     sg.write('*RST')
55     sg.write(f':SOUR:POW {input_power_dbm}')
56     sg.write(':OUTP ON') # Turn RF ON
57
58     print("Starting Frequency Sweep...")
59
60     for freq in frequencies:
61         # Set SG frequency
62         sg.write(f':SOUR:FREQ {freq}')
63
64         # Set SA center frequency to match SG
65         sa.write(f':SENS:FREQ:CENT {freq}')
66
67         time.sleep(0.1) # Wait for instruments to settle
68
69         # Trigger SA sweep
70         sa.write(':INIT:IMM')
71         sa.query('*OPC?') # Wait for sweep to finish
72
73         # Find peak (Marker to maximum)
74         sa.write(':CALC:MARK1:MAX')
75         output_power_str = sa.query(':CALC:MARK1:Y?')
76         output_power_dbm = float(output_power_str)
77
78         # Calculate scalar Gain
79         gain = output_power_dbm - input_power_dbm
80         gains.append(gain)
81
82         print(f"Freq: {freq/1e9:.3f} GHz | Gain: {gain:.2f} dB")
83
84     # Turn off SG output for safety
85     sg.write(':OUTP OFF')
86
87     # Plot Results
88     plt.figure(figsize=(10, 6))
89     plt.plot(frequencies / 1e9, gains, color='red', linewidth=2,
marker='o')
90     plt.title('LNA Scalar Gain vs Frequency (SG + SA Method)')
91     plt.xlabel('Frequency (GHz)')
92     plt.ylabel('Gain (dB)')
93     plt.grid(True, which='both', linestyle='--', alpha=0.7)
94     plt.axhline(0, color='black', linewidth=1)
95     plt.show()
96
97     except Exception as e:
98         print(f"An error occurred: {e}")
99     finally:
100         if 'sg' in locals(): sg.close()
101         if 'sa' in locals(): sa.close()
102         print("Connections closed.")
103
104 if __name__ == '__main__':
105     run_sg_sa_test()

```